

Insurance Prediction

- 1. Problem Statement:** Predict insurance charges based on patient parameters using a machine learning regression model.
- 2. Dataset Info:** 1338 rows and 6 columns (age, sex, bmi, children, smoker, charges).
- 3. Pre-processing:** One-Hot Encoding was used to convert categorical string data (sex and smoker) into numerical data, along with standard scaling for the SVM model.
- 4. Model Development:** Multiple Linear Regression, Support Vector Machine (SVM), Decision Tree, and Random Forest algorithms were developed and evaluated.

5. Research Values (R^2 Scores):

1) Multiple Linear Regression – R^2 Value = 0.7894

2) Support Vector Machine

S.No.	Hyper Parameter	Linear (r value)	RBF – Non-Linear (r value)	Poly (r value)	Sigmoid (r value)
1	C10	0.4625	-0.0323	0.0387	0.0393
2	C50	0.6093	0.1478	0.4152	0.4047
3	C100	0.6289	0.3200	0.6180	0.5276
4	C500	0.7631	0.6643	0.8264	0.4446
5	C1000	0.7649	0.8102	0.8566	0.2875
6	C2000	0.7440	0.8548	0.8606	-0.5940

The **Support Vector** Regression's Highest R^2 Value (C2000,Poly) is = 0.8606

3) Decision Tree

S.No.	Criterion	Max Features	Splitter	r Value
1	Poisson	None	Best	0.7301
2	Poisson	None	Random	0.6489
3	Poisson	Sqrt	Best	0.7079
4	Poisson	Sqrt	Random	0.6552
5	Poisson	Log2	Best	0.7079
6	Poisson	Log2	Random	0.6552
7	Absolute_Error	None	best	0.6530
8	Absolute_Error	None	random	0.7187
9	Absolute_Error	Sqrt	Best	0.7668
10	Absolute_Error	sqrt	Random	0.7921
11	Absolute_Error	log2	Best	0.7668
12	Absolute_Error	Log2	Random	0.7921
13	Squared_Error	None	Best	0.6798
14	Squared_Error	None	Random	0.7256
15	Squared_Error	Sqrt	Best	0.7277
16	Squared_Error	Sqrt	Random	0.6715
17	Squared_Error	Log2	Best	0.7277
18	Squared_Error	Log2	Random	0.6715

The **Decision Tree** Regression's Highest R^2 Value (MAE,sqrt&log2,random) is = 0.7921

4) Random Forest

S.No.	Criterion	Max Features	N_estimators	r Value
1	Poisson	None	100	0.8556
2	Poisson	Sqrt	100	0.8715
3	Poisson	Log2	100	0.8715
4	Absolute_Error	None	100	0.8478
5	Absolute_Error	Sqrt	100	0.8706
6	Absolute_Error	Log2	100	0.8706
7	Squared_Error	None	100	0.8542
8	Squared_Error	Sqrt	100	0.8735
9	Squared_Error	Log2	100	0.8735

The **Random Forest** Regression's Highest R^2 Value (MSE,sqrt,100) is = 0.8735

6. Final Model & Justification: The **Random Forest** (MSE,sqrt,100) is the final model because it achieved the highest overall R^2 score of 0.8735. It provides an excellent balance between making the trees diverse (uncorrelated) and ensuring each individual tree still has enough good features to make accurate predictions.